

Efficiency of Panopto

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Abstract—The objective of this research paper is to investigate the efficiency of Panopto as an e-learning tool. Due to the increasing use of remote learning, Panopto has emerged as a popular platform for recording and sharing educational lectures. However, the platform's efficiency for students depends on its usability, which affects the accessibility and consumption of course material. Therefore, this study aims to identify where there is a decrease in efficiency and how this could affect user experience. The research question is "How efficient is Panopto as an e-learning tool?" The paper covers the background of efficiency evaluation and the features of Panopto. The research method selected is an experiment, the data collection method is observation and the data analysis method is quantitative. The study is intended to assist designers in making better video-sharing platforms by identifying patterns and areas that could use improvement. Based on the results of our study, it can be concluded that the platform's efficiency is flawed, as the students' comprehension level aligns with our experiment hypothesis. Hence, the findings serve as a response to the research question.

Index Terms—Panopto, usability, efficiency

I. INTRODUCTION

With the rise of remote learning, students are often required to access video lectures on their own time and from various locations. One of the platforms used for this is Panopto, which has become popular for recording and sharing educational lectures, making it a valuable tool for students. Panopto allows you to create recordings directly from your computer, record lectures in certain classrooms, and upload recordings through an integrated playback platform. Panopto is integrated with Blackboard, and recordings can be published to students easily. This is how Panopto is described on the NTNU website [1].

There already exists research on the effectiveness and efficiency of web-based e-learning tools in general, where effectiveness is defined in terms of cost, students' acceptable levels, and performance for educational strategies. [2] For the purpose of this study, we have defined efficiency as the ability to complete a task in a way that wastes the least amount of time, effort, and resources. The research objective is to evaluate the efficiency of this video-sharing tool and detect the problems that come with it. By researching the efficiency of Panopto as a platform, one can identify ways to improve its accessibility, responsiveness, and overall user experience, making it easier and more efficient for students to watch lectures and engage with course content. Furthermore, improved efficiency can have a positive impact on student's academic success. In this study by the University of Wales, the

researchers concluded that Panopto enabled students to revisit and access material outside of the classroom, which provided more flexible and adaptable learning opportunities than that provided by traditional teaching [3]. Panopto's improved video lecture accessibility and engagement can benefit academic performance by allowing students to focus on learning, while also benefiting educators and instructional designers by identifying areas for improvement in course content delivery. This is the motivation for our research question:

RQ: How efficient is Panopto as an e-learning tool?

II. BACKGROUND

The concept of efficiency was brought up in the introduction, it has been defined as how time efficient it is, how many resources are necessary, and how little effort the user needs to access information. Comparatively, there exists research regarding Panopto, but it only discusses the benefits of the platform. This research found that the use of Panopto complemented classroom teaching, and was beneficial for the majority of the students [3]. Similarly, another research paper regarding "Estone", which is another platform that provides video lecture capture for educational institutions, discusses and analyses the benefits of switching to the platform, but not the drawbacks [4]. Due to the insufficient data on the user's interaction with the platform in previous research papers, we decided to add relevant data to this field.

Based on the concept of efficiency we decided to test user satisfaction by testing the time they used to complete a task, if they need assistance to find information, how the user expects to be able to find their resource, the effort of which they used to search or the effort to remember steps needed to proceed.

To test the participants when using the platform we decided to use a usability testing method described in a paper by E. Wang & B. Caldwell [5]. Here they compare two different usability-testing methods, heuristic evaluation, and user testing. There are different strengths and weaknesses of both methods. "User Testing can provide more insightful data from real users such as user's performance and satisfaction." The paper also specifies user testing as more appropriate for a product later in the development cycle, where "User Testing should be considered as a major testing method because it can capture real usability problems from the real users." Based on this, user testing was chosen as the optimal testing method for this study.

III. METHODS

A. Research methods

Since the purpose of the study is to understand the efficiency of Panopto, conducting an experiment is the research strategy that best suits this task. With an experiment, there needs to be a hypothesis. For this research, the hypothesis is that Panopto's design is not focused on user efficiency and users will, in return, spend a lot of time on the tasks. When using a video-sharing service, participants want to be able to find what they are looking for right away, optimally with a quick search function if possible. This should not take more than 25 seconds and therefore this is the baseline that is used in the experiments. This number is based on the general time a user would type in their search query and identify their target.

The goal of an experiment is to investigate cause and effect. In our case, we study whether the design has an effect on the time participants spend when performing tasks, as well as trying to prove or disprove the hypothesis through observations or measurements. The time measurements are taken to test various parts of Panopto's design by performing a variety of tasks. [6]

Other research methods include case studies and survey-based approaches. With a case study, the data obtained is to be qualitative as it should give the researchers a deeper understanding of the case and the complex relationships between processes. As this case focuses on a simple and numeric part of the process, we didn't need to perform a case study. With a survey, the goal is to gather the same data from a large number of people. Even with the goal being to gather quantitative data and look for patterns, surveys often rely on questionnaires as their data-generating method, while case studies rely on observations. Observation is the data-generating method that is used. [6]

B. Data generation methods

As mentioned, the data-generating method to be used is observations. This is due to the nature of this experiment. The goal is to observe the amount of time participants take. This can only be done through observations as holding interviews or sending out questionnaires would not gather the desired data. Therefore, it is possible to rule out research strategies that use these types of data-generating methods such as surveys and design and creation. This is due to the nature of trying to understand efficiency. To measure efficiency, the subjects are given several tasks to complete. By measuring how quickly they complete these tasks, we can calculate the time-based efficiency of Panopto [7].

C. Data analysis method

The data that is generated through the experiment is quantitative, which is therefore the data analysis method. As the goal of the study is to test efficiency, there is a number needed to identify how efficient systems are [6]. Through this data, the objective is to see patterns and areas where there are problems. With quantitative data analysis, there is to be a focus on the statistical aspect. The data collected is ratio

data, as there is a true zero. The starting time will always be 0 and, therefore, the data can be compared with addition, subtraction, multiplication, and division. The data will be represented in a table, making it easy to compare test subjects to each other and get an overall overview of the total results. When working with statistical analysis, specifically inferential statistics, it makes the numbers and research easier to digest. With inferential statistics using methods like hypothesis testing can help give a generalized approximation for the users of the platform. The main threat to testing efficiency is how well the subjects know the system beforehand, as this system has been in use for quite some time at NTNU some subjects could have already frequently used it. This is part of the problem with the experiment, but since all test subjects have seen the system and used it before, there is a sort of baseline for the participants. Still, even with that being the case participants can struggle with tasks that may be outside of their use area for the website.

D. Participants

Our research focuses on an e-learning tool for high-level students. Access is limited to the viewer version of Panopto, so participants are restricted. The experiment involves our group as facilitators and observers and a group of student subjects. Technological abilities do not affect the goal of testing efficiency, so students from any background can participate. Different types of subjects are necessary to test the impact of technological backgrounds on understanding websites.

Although the experiment is narrow in scope, its general aspects can be applied to making efficient video platforms for lectures. Panopto's user interface team or other video providers can use this research.

As facilitators, our role is to observe, not assist, the subjects while they complete tasks. Mistakes and completion time are relevant to the total task time, so we must not interfere. Leaked tasks could result in subjects practicing and performing better. Tasks must be general to test Panopto's different abilities. We must be unbiased and neutral to avoid influencing the subjects.

To accurately time task completion, we will use a stopwatch to record the data. Anonymity and confidentiality are not of concern since we only need to observe the process, without a need to gather personal data. We will provide computers to ensure the all participants share the same experience without deviation. [6]

E. How the Experiment is Conducted

The experiment will be held in a room with a computer set up to Panopto's front page as that is the starting point of every task. The participants in the experiment are at least two researchers, one being the facilitator and the other being an observer, and a participant being the test subject. Before the experiment starts the facilitator will inform the subject of their right to withdraw at any moment and that all information will be anonymous. The facilitator will have a stopwatch where they record the time it takes for each subject to complete the tasks and write down this time. Before the experiment starts the facilitator asks the participants "How would you rate your

technology skills on a scale from 1 to 5?”, this could help give an understanding of the different times that participants have. Having good technical skills can help when performing tasks on a computer. The facilitator will ask the subject to complete the following five tasks.

- 1) Find the first recording from the lecture in the subject TMA4100 from fall 2022. (TMA4100 Matematikk 1 - Informasjon om Matematikk 1 (del 1))
- 2) Find the second recording from the lecture in the subject TMA4100 from fall 2022. (oversiktsforelesning 1)
- 3) You have been sick and missed the lectures from week 44, find them in the subject TMA4100 from fall 2022. (This is from the 31 of October to 6 of November) (oversiktsforelesning 10)
- 4) Find and subscribe to NTNU Trening. From subscriptions find the longest yoga session from NTNU Trening (yoga 31. March)
- 5) Unsubscribe from NTNU trening

By having subjects go through these questions, we test the different abilities that Panopto has to offer. As the goal is to test efficiency, there have to be generic tests of the different tools to test how efficient they are. The goal of the experiment is to gather quantitative data, where the focus is on the amount of time tasks took. After all the data is collected from the participants, it will be gathered and analyzed to see what can be concluded from the results.

IV. RESULTS

The experiment gathered time measurements from a total of 35 tests from 7 participants. All of the participants were students with knowledge about the platform, but with varying amounts of experience. The group decided to separate the participants based on their perceived technical experience on a scale from 1 to 5.

The results from Table I show that the participants used on average more than our baseline for acceptable time usage on tasks 1, 2, and 4. With task 4 being the outlier with the highest average. Approximately 71% of the participants struggled with the first task, 80% of them ranking themselves as 4/5 on the competence scale. Similarly, 43% struggled with task 2, 29% on task 3, 100% on task 4, and 14% for task 5.

What stood out was the fact that both groups of experience struggled with the same amount of tasks, but not the same ones. Those with 4/5 experience struggled with 1, 2, and 4 and those with 5/5 experience struggled with 2, 3, and 4, but to a lesser degree on everything except the fourth.

When performing the hypothesis test, there needed to be a hypothesis zero. The hypothesis stated that users would spend more than 25 seconds, so our hypothesis zero (H_0) is that; tasks will take more time than 25 seconds with a standard deviation of 5 seconds, which is 20% of the time. To calculate if the hypothesis is rejected or accepted, there would need to be a hypothesis α (H_α) stating that the time is less than 25 seconds. Using the formula $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ with x being the mean time used, μ being the hypothesized time, σ being the standard

TABLE I

SHOWS THE TIME USERS USED ON THE DIFFERENT TASKS GIVEN TO THEM. THE TASKS CAN BE FOUND IN SECTION III-E. ALONG WITH HOW THE AVERAGE OF THE SUBJECTS COMPARE TO THE HYPOTHESIS OF 20 SECONDS.

Subject	Task 1	Task 2	Task 3	Task 4	Task 5
Student 1 (5/5)	00:23.30	00:24.00	00:10.00	1:42.50	00:15.00
Student 2 (5/5)	00:09.00	01:21:10	00:50.00	02:35.00	00:13.00
Student 3 (4/5)	01:41.50	00:15.00	00:20.00	05:24.00	00:36.50
Student 4 (4/5)	00:19.20	01:26.20	00:26.00	01:06.20	00:13.50
Student 5 (4/5)	02:00.00	00:45.00	00:20.00	02:06.00	00:11.00
Student 6 (4/5)	02:01.00	00:13.46	00:22.31	02:03.51	00:09.06
Student 7 (4/5)	00:59.35	00:14.03	00:15.21	01:58.67	00:10.71
5/5 experience	00:16.15	00:52.55	00:30.00	2:08.75	00:14.00
4/5 experience	01:14.21	00:34.73	00:20.70	2:31.67	00:16.15
Students	00:59.19	00:39.80	00:23.36	02:15.12	00:15.04
Difference from baseline	00:34.19	00:14.80	-00:02.36	01:50.12	-00:10.04

deviation and n being number of subjects. From this, a p-value was calculated and the conclusion was that for tasks 1, 2, and 4 H_0 could not be rejected. This is in line with the original hypothesis that tasks would take longer than necessary.

The results indicated that, on average, the participants' time usage exceeded our baseline. As this was the baseline for our concept of efficiency the results answer the research question.

V. DISCUSSION

When looking at the results, we see that subjects that defined their experience as 5/5 performed better on average than the subjects that defined their experience as 4/5. On task 4, for example, one of the subjects with 4/5 experience needed more than five minutes to complete the task. In comparison, the highest time usage among 5/5 experience subjects was only two and a half minutes. This could be attributed to the fact that the platform's usability is not the best. The RQ of this research focuses on the efficiency aspect of usability, where we have defined efficiency by time usage, resources, and effort needed to solve a task. With poor efficiency, those with lower technical skills can struggle to find what they need in a reasonable time. This could cause users to become frustrated and want to give up and not continue using the website. Proper design is important for a website to keep and grow the user base [8].

Based on the findings, we can compare how the findings match our hypothesis; "Panopto's design is not focused on user efficiency and users will, therefore, spend a lot of time on the tasks.". Out of the five tasks, for three of them, we could not reject this hypothesis. Some of them, like task 4, are quite far outside the expected time range. However, for the other two tasks, the average time was lower than the expected baseline. Showing that different tasks asking users to test out different aspects are designed differently. One could argue about how the contents of each task are different, and therefore the time needed to complete a task would also differ. This can, however, not be the only reason for the drastically long times for tasks 1 and 4. The design of the website should not allow for some tasks to take a long time, while others can be done easily.

By looking at Table I, it can be seen how the test result and the hypothesis do not match up. To prove this mathematically, inferential statistical analysis was done. To achieve this, a hypothesis test was performed. Hypothesis H_0 , stating that the

subject would use more than the expected time, and hypothesis H_a , stating that the subject would spend less than the expected time. This resulted in tasks 1,2 and 4 proving H_0 , meaning that there is mathematical evidence to show the hypothesis that was created is valid. Through both mathematical calculations and the visible evidence, the efficiency in Panopto is lacking, causing it to be a sub-optimal tool for e-learning.

Panopto hosts a vast library of video content, which can include thousands of lectures, presentations, and other types of educational content. Finding a specific lecture within this vast library can be challenging, especially if the user is not familiar with the organization of the content or does not have specific search criteria to narrow down their search. Thus, a 25-second baseline might not be realistic, and it may take longer than 25 seconds to find a lecture on Panopto, especially for users who are new to the platform. To accurately analyze the results of our experiment, we should consider factors such as the user's familiarity with the website, their experience using similar platforms, and the complexity of the search criteria required to find the lecture. We should also allow sufficient time for users to complete the task, even if it takes longer than 25 seconds. If previous experience with the platform was noted down before the experiment, this factor could be reduced.

There were also some complications with the tasks defined for the experiment. One example is task 3, which asked the user to find the lectures from week 44, 2022. However, more than one lecture from week 44 was available. This caused different participants to find different lectures from that week. In real life, if a user were sick for a whole week they would need to find the first lecture of that week and work their way from there, but as this was not specified, some individuals found themselves feeling like they had completed the tasks before they had found the lecture that we had anticipated they should find. To combat this, the task should have been more specific and well-defined, so that the user knew exactly what they were expected to do. This along with other complications made it so that the data obtained on average shows less time than the tasks would have taken if they knew specifically what they were looking for.

It is interesting to see that despite the findings from the research showing Panopto as a flawed tool, it still has a positive impact on the students' learning [3]. If the tool could be improved based on our findings, we believe that this impact would be even greater.

VI. CONCLUSIONS

Panopto works in general for sharing educational lectures, but is not optimized in regards to time usage on the website, and has issues being used by itself instead of through other sources. With a focus on its usefulness, we delved deeper into understanding the usability of the platform and how that can be measured. We chose an experiment that would be more beneficial for the results, which were gathered by observing the participants finishing our set tasks. Observing the results we can see that technically experienced participants struggle to complete what we imagined as normal situations a user

might encounter when using the website. By optimizing this interaction and focusing on improving the flaws we focused on, in return we believe the usability of the website will increase, leading to a higher satisfaction rate for all students and educators alike.

However, there were some limitations in our research. The wording of the tasks was not always clear to the subject, which could negatively impact the findings. It is unclear if the baseline set for the hypothesis is accurate or not. This is something that could be improved to get a more accurate result. Including the subjects' previous experience of the platform, and incorporating it into the baseline could also help increase the accuracy of our findings.

Further research would be to use this data and try to improve upon the existing system. This research is very specific for Panopto as a website, and it would therefore be most useful for Panopto to continue the work of improving its usability. However, this can be looked at in a general way as well. Since the tasks that were chosen for testing were generic in how such a system should work, the goal for other video-sharing platforms should be to have systems that are more efficient when trying to find videos and such. Overall, this study could be used by anyone, but there would need to be more usability testing to make sure changes to the system give users a better experience.

REFERENCES

- [1] Section for Teaching and Learning Support. "Panopto lecture capturing". <https://i.ntnu.no/wiki/-/wiki/English/Panopto+lecture+capturing>. Accessed, 2023.
- [2] Wai, C. C., & Seng, E. L. K. (2014). Exploring the effectiveness and efficiency of blended learning tools in a school of business. *Procedia-social and behavioral sciences*, 123, 470-476.
- [3] Bowes, K., Dasari, S., & Hall, J. (2017, July). Assessment of Panopto software for teaching undergraduate anaesthesia in Namibia. In *ANAESTHESIA* (Vol. 72, pp. 19-19). 111 RIVER ST, HOBOKEN 07030-5774, NJ USA: WILEY.
- [4] Qingsheng, L., Jipeng, W., & Zhang, R. (2010, July). Research and design of teaching video recording and broadcasting system. In *2010 3rd International Conference on Computer Science and Information Technology* (Vol. 4, pp. 45-50). IEEE.
- [5] Wang, E. & Caldwell, B.S. "An Empirical Study of Usability Testing: Heuristic Evaluation Vs. User Testing" ResearchGate . Accessed Mar. 2023, published, sept. 2002.
- [6] Oates, Briony J., Marie Griffiths, & Rachel McLean. *Researching information systems and computing*. Sage, 2022.
- [7] Arthana, I. K. R., I. M. A. Pradnyana, & G. R. Dantes. "Usability testing on website wadaya based on ISO 9241-11." *Journal of Physics: Conference Series*. Vol. 1165. No. 1. IOP Publishing, 2019.
- [8] Kuan, H. H., Bock, G. W., & Vathanophas, V. (2005, January). Comparing the effects of usability on customer conversion and retention at e-commerce websites. In *Proceedings of the 38th annual Hawaii international conference on system sciences* (pp. 174a-174a). IEEE.

VII. REVISION TABLE

#	Reviewer comment	Response	Changes done to
1 <i>abstract</i>	Talk about the summary of your findings in the Abstract	We have understood that this was a important part missing.	Changed the abstract to include it.
2	Explain how your results answer to research question briefly in the Abstract	Agreed that this was an important part missing.	Including a correlation between the results and the RQ.
3 <i>intro</i>	What is the research objective? Do you want to present a model for enhancing the usability of Panopto, or do you want to evaluate the Tool's efficiency and find the problems? Please define your goals and objectives in the introduction clearly.	We understand we hadn't been clear on this issue.	Specify the objective and the definition of our goal.
4	What does efficiency mean in your research? What factors contribute to it? Try to address related studies and how they evaluate the efficiency of video recording or similar educational tools.	We agree that these are important aspects to discuss and mention.	Our definition of efficiency and added two related studies describing efficiency and the benefits of Panopto.
5	Your objective of the study seems biased as the explanation includes solutions for your problem.	We agree that it was biased and have removed it.	Remove and rewrite parts of the introduction.
6	Your research question needs to be formulated to address your objective. From the research question it seems that you just wanted to know if Panopto is efficient or not.	As we've defined efficiency, we do not see the issue of keeping the RQ as is.	No changes.
7 <i>background</i>	The research topic concerns the efficiency of video recording tools in the education system, but the background section focuses more on usability. Besides defining the efficiency concept, address related studies previously evaluated using similar educational tools, their framework, and their findings.	We understand it is beneficial for the research to relate to another study, thus we added comparison studies to show the gap of knowledge we wanted to fill.	Refocused the background with regards to efficiency, added two related studies and compared their findings.
8	After describing related studies, analyze how these studies relate to each other. What is the research direction, and how they addressed the topic? You can also use these data in the discussion section to analyze your data.	Connected the research papers by comparison. They are analyzed, but not connected to other parts of the paper, as we wanted to fill the insufficient data we couldn't find.	Used the research focus and findings in the comparison.
9 <i>method</i>	The research strategy that you have described is an experiment, not a case study. You have designed an experiment to evaluate the Tool's efficiency. Therefore, explain the different aspects of the investigation.	We agree that this is an experiment.	Updated the research methods from case study to experiment and changed the method reasoning.

10	The statement about choosing the research strategy must be stronger and more explained. Focus more on why the experiment is more suitable than a survey or design and create and explain the advantages of using this strategy to evaluate efficiency.	We agree that we need to have a stronger explanation for why experiment is the best fit.	Updated the explanation as to why survey would not fit this type of research.
11	Explain the data analysis approach. Recommend to formulate a hypothesis and use quantitative analysis and inferential statistic tests to analyze the data. You have enough data but should use a data analysis approach to extract the results.	We agree that the data analysis approach should be presented more clearly.	Updated the method section to include an explanation of our approach, how we specify and use the data and usage of the hypothesis in the inferential statistics.
12	Describe every participant's experience and how they react to tasks	The goal of the study is efficiency and not user perception, therefore, the user experience is used in discussion.	No changes.
13	Review the chapter for experiment in the Textbook. You need to know how to report an experiment.	As the methodology was changed, we rewrote the research strategy.	rewrote sections of the methodology and results.
14	The analysis method must be Qualitative.	As we've described the difference in benefits between qualitative and quantitative we do not agree with changing the analysis method.	No changes.
15 <i>result</i>	Return to your research question and address the user reaction through the Lens of your RQ. You have said," We found that students with lower competence levels had a steeper learning curve when advancing from	We understand our results weren't focused on the right approach and adapted to include this.	Rewrote results section with a focus solely on analysis of data and relevance to the research question.
16	Show claims through data by explaining it from a user perspective. We talk more about the data, not our opinion in Result section.	Agree, that the focus should solely be on data.	Rewrote results with focus on analysing and explaining data.
17	Refer to "participants" as "Student 1", "Student 2", etc... for more clarity and readability.	We don't feel the need to point out individual results, as we aimed to generalize the participants.	No changes.
18 <i>discussion</i>	Categorize the findings and relate them to your research question, as the study lacks a conceptual model in the discussion. It is hard for authors to connect the results to theories and background studies. As said in the background section, find related studies and define your model before analyzing data in this section.	We agree that this would improve the paper.	Improved background section and restructured discussion section to relate to conceptual model.

19	Describe efficiency in terms of a model that consists of factors in the Background section. Then in the discussion, you can talk about each factor and then give an overview of the data from different perspectives	We agree that the discussion part lacked structure.	Restructuring.
20	Use the reference literature from the background section and compare your findings with other scholars' findings in this section. Try to refer to those studies and then highlight your contribution compared to others.	Previous background literature is used to justify, but does not correlate with findings.	Mentioned findings from background literature as a comparison to our contribution.
21	Add subsections to make the findings more readable.	Don't see the need to separate the discussion.	No changes.
22 <i>conclusion</i>	Talk about limitations and future studies in the conclusion section.	Limitations were missing and future studies was unclear.	Added limitations and changed wording on future studies.
23	The conclusion is formulated well but should include clear findings of your research. Also include some suggestions for the improvement of Panopto as your study aims for it.	Our study aims to show how efficient Panopto is as a resource and therefore do not feel the need for including suggestions.	No changes.
24 <i>no field</i>	You have used the words "efficiency" and "effectiveness" inconsistently.	We agree that we had made some conceptual mistakes.	Rewrote the paper to focus on efficiency only.
25	References should be numbered correctly.	We agree.	Fixed references.